

2.7 Multiplying Binomials and Polynomials

Lesson Introduction

 Benchmark	MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.		 Learning Targets	- I can multiply binomials. - I can multiply polynomials.
 Benchmark Coherence	Building On MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.		Working Toward MA.912.AR.1.5 Divide polynomial expressions using long division, synthetic division, or algebraic manipulation. MA.912.AR.1.9 Apply previous understanding of rational number operations to add, subtract, multiply, and divide rational algebraic expressions.	
 Essential Questions	- How is the distributive property associated with multiplying polynomials? - How can you multiply polynomials?			
 Lesson Narrative	In this lesson, students extend their learning from the prior lesson to being able to multiply polynomials with different numbers of terms. Students will begin by connecting multiplying binomials to the distributive property using the area model. They will then collaborate to apply their learning to multiplying polynomials with more than two terms.			
 Component Summary	2.7.1 Warm-Up (~5 min.)	Notice and Wonder about a dot plot (<i>SE p. 92</i>)	2.7.4 Collaboration (10 min.)	Extend to multiplying polynomials using the distributive property and the area model (<i>SE p. 94</i>)
	2.7.2 Guided Instruction (10-15 min.)	Connect representations of multiplying binomials to the distributive property (<i>SE p. 92</i>)	2.7.5 Your Turn! (10 min.)	Finding the product of two polynomials (<i>SE p. 95</i>)
	2.7.3 Guided Practice (5 min.)	Practice multiplying binomials (<i>SE p. 94</i>)	2.7.6 Wrap-Up (~5 min.)	Reflection on understanding of the lesson (<i>SE p. 96</i>)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Binomials - Polynomials	(Optional) Anchor Charts (Optional) Algebra tiles (Optional) Highlighters	- Work through each of these problems prior to facilitating with students to be able to easily spot errors. Students often make minor calculation errors and missteps. - For those students who prefer the area model method, consider providing blank area models of different sizes that students can use when multiplying.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may fail to apply the distribution to all terms from one polynomial to another. - Students may multiply the signs incorrectly when distributing. - Students may forget to apply the proper laws of exponents when multiplying.	- This lesson leverages the use of the distributive property and connecting multiple representations of multiplying polynomials to this property. Students may need a very brief refresher of the distributive property and the misconception of not multiplying all terms by the distributed factor. - Allowing students to use scientific calculators during the lesson will help students focus on discovery and process of multiplying polynomials. Consider having no-calculator questions where students turn their calculator over.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 2.7.1 Warm-Up engages students in a Notice and Wonder protocol. Students will describe what they notice when looking at the dot plot and what they wonder about the data presented.	MA.K12.MTR.2.1: Multiple Representations Throughout the lesson, students will explore multiplying polynomials using various methods and representations. These include the distributive property, algebra tiles, and area models. This provides students opportunities to use manipulatives when investigating concepts. Through this lesson progression, teachers can guide students from concrete to pictorial to abstract representations as their understanding progresses.
 Differentiate Instruction	Students often struggle with the process of multiplying polynomials when it comes to the accurate computation and the steps. Consider using colored highlighters or markers to highlight what happens in each step. This can help students visualize the process during the distribution of each term.	
	Consider providing an Anchor Chart that shows the multiplication of different types of polynomials, notating any key differences that students can refer back to throughout the lesson and unit.	
Lesson Notes:		

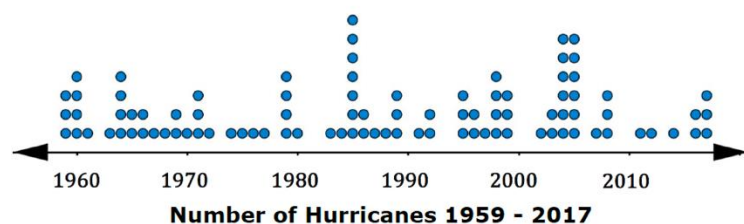
2.7.1 Warm-Up: Notice and Wonder

Component Overview: Students will engage in a Notice and Wonder activity with a dot plot presenting data related to hurricanes.



2.7.1 Warm-Up: Notice and Wonder

- The dot plot shows the number of hurricanes that made landfall in the continental United States from 1959 – 2017.



- What do you notice when looking at the dot plot?

Answers will vary. I notice that the most hurricanes to make landfall happened in 1985. I notice that there are more dots between 1980 – 1990 and between 2000 – 2010.

- What do you wonder about the data presented?

Answers will vary. I wonder if there were any major environmental changes during the mid 80's and 2000's. I wonder why there weren't as many spikes in data in the mid 70's.



This activity is richer when the students get to talk about their answers to question 1a and 1b. Students will often verbalize more than they will write.

Consider converting these questions to discussion questions with a partner or having them share their answers out as a whole group.

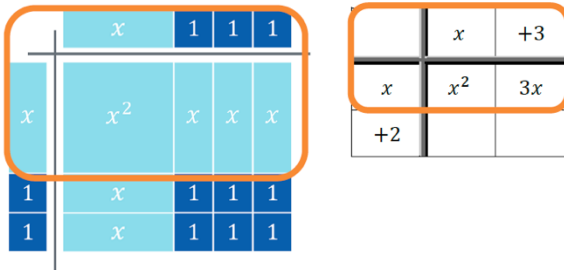
2.7.2 Guided Instruction: Multiplying Binomials

Component Overview: Students will make connections between using algebra tiles and the area model to the distributive property.



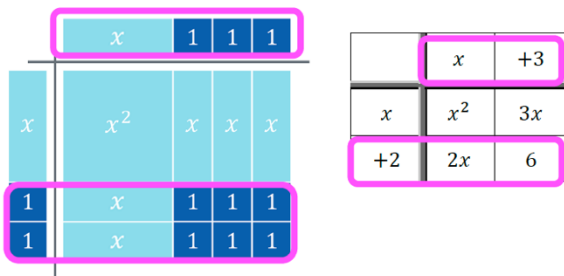
2.7.2 Guided Instruction: Multiplying Binomials

An algebra tiles model and an area model of $(x + 2)(x + 3)$ are shown. The orange box in both models shows that the x from the first factor was distributed to the $(x + 3)$.



Model explicit connections between the models used to ensure students to better connect the concepts and representations.

The pink boxes show the 2 from $(x + 2)$ being distributed to the $(x + 3)$.



After students have had time to read and analyze these models, consider asking the following:

- How are the numbers in the area model related to the model using algebra tiles?

2.7.2 Guided Instruction: Multiplying Binomials (continued)

Distributing the x and the 2 from $(x + 2)$ to $(x + 3)$ can be algebraically represented as $x(x + 3) + 2(x + 3)$.

1. Apply the distributive property. Then combine like terms.

$$x(x + 3) + 2(x + 3)$$

$$x^2 + 3x + 2x + 6$$

$$x^2 + 5x + 6$$

2. Discuss with your partner how the algebra tile model or area model relates to applying the distributive property.

Answers will vary. The algebra tile model relates to distributive property because you are multiplying the tiles representing the height times the tiles representing the width. The area model relates to applying the distributive property because you are multiplying two terms from two different factors, creating partial products that will combine to create the final answer.



Model how to apply the distributive property including thinking aloud and marking the expression as students work through it with the whole group.



Consider employing a Think-Pair-Share protocol for question 2. Allow students a minute to think about it on their own and then pair up with their partner to discuss before having them share out as a whole group.

2.7.2 Guided Instruction: Multiplying Binomials (continued)

Brayden and Jackson were asked to find the product of $(w - 8)(w + 5)$ using different methods.

- Complete the work for each student using the method they were assigned.

Brayden's Distributive Method	Jackson's Area Model									
$w(w + 5) - 8(w + 5)$ $w^2 + 5w - 8w - 40$ $w^2 - 3w - 40$	<table><tr><td></td><td>w</td><td>$+5$</td></tr><tr><td>w</td><td>w^2</td><td>$5w$</td></tr><tr><td>-8</td><td>$-8w$</td><td>-40</td></tr></table> $w^2 - 3w - 40$		w	$+5$	w	w^2	$5w$	-8	$-8w$	-40
	w	$+5$								
w	w^2	$5w$								
-8	$-8w$	-40								

- Multiply $(m^3 + 2k)(5m - 6k)$ using the distributive property.

$$m^3(5m - 6k) + 2k(5m - 6k)$$

$$5m^4 - 6km^3 + 10km - 12k^2$$

- Find the product $(2a + 3b)^2$ using an area model.

	$2a$	$+3b$
$2a$	$4a^2$	$6ab$
$+3b$	$6ab$	$9b^2$

$$4a^2 + 12ab + 9b^2$$



Allow students to share their answers, along with their process, for each method. Add a layer of engagement by having students vote for which method they find easier, providing rationale for why they find that way easier.

Consider splitting the class in half and having each half use a different method to find the product before discussing as a whole group.



Model how to multiply two binomials using the distributive property.



Consider asking students to set up the area model on their own and then work with a partner to find the product.

2.7.3 Guided Practice: Multiplying Binomials

Component Overview: Students will complete two practice questions with multiplying binomials using their method of choice.



2.7.3 Guided Practice: Multiplying Binomials

Find each product using your preferred method. Show your work.

1. $(17 - x)(1.2x^2 + 6)$

$$-1.2x^3 + 20.4x^2 - 6x + 102$$



Consider allowing students to use a calculator so they can focus on the process rather than the computations at this point in instruction.

2. $\left(\frac{1}{4}m + 13n\right)^2$

$$\left(\frac{1}{16}m^2 + \frac{13}{2}mn + 169n^2\right)$$

2.7.4 Collaboration: Multiplying Polynomials

Component Overview: Students work together to multiply polynomials with different numbers of terms using the area model and the distributive property. For this activity, students should work together in partner pairs or small groups.



2.7.4 Collaboration: Multiplying Polynomials

The same methods can be extended to multiply any polynomial.

1. Work with your partner to complete the following.

- a. Multiply $(y - 1)(y^2 + 7x - 26)$ using the area model.

	y^2	$+7x$	-26
y	y^3	$7xy$	$-26y$
-1	$-y^2$	$-7x$	26

$$(y - 1)(y^2 + 7x - 26) = y^3 + 7xy - y^2 - 7x - 26y + 26$$

- b. Multiply $(1.5x - 5)(4x^2 - 7x + 3)$ using the distributive property.

$$1.5x(4x^2 - 7x + 3) - 5(4x^2 - 7x + 3)$$

$$6x^3 - 30.5x^2 + 39.5x - 15$$

- c. Compare your answers with your partner and resolve any differences.

Answers will vary.



For this activity, provide students with the general logistic directions and release them to try and complete the entire activity with their partner or small group. Monitor the groups, asking clarifying questions when observing.

2.7.4 Collaboration: Multiplying Polynomials (continued)

2. Decide who will be Partner A and who will be Partner B.
- a. Find the product of $(2n^5 - n - 14)(3n + 2)$ using the method indicated.

Partner A Distributive Property	Partner B Area Model																
$(2n^5 - n - 14)(3n + 2)$ $3n(2n^5 - n - 14) + 2(2n^5 - n - 14)$ $6n^6 - 3n^2 - 42n + 4n^5 - 2n - 28$ $6n^6 + 4n^5 - 3n^2 - 44n - 28$	<table><tr><td></td><td>$2n^5$</td><td>$-n$</td><td>-14</td></tr><tr><td>$3n$</td><td>$6n^6$</td><td>$-3n^2$</td><td>$-42n$</td></tr><tr><td>2</td><td>$4n^5$</td><td>$-2n$</td><td>-28</td></tr><tr><td colspan="4">$6n^6 + 4n^5 - 3n^2 - 44n - 28$</td></tr></table>		$2n^5$	$-n$	-14	$3n$	$6n^6$	$-3n^2$	$-42n$	2	$4n^5$	$-2n$	-28	$6n^6 + 4n^5 - 3n^2 - 44n - 28$			
	$2n^5$	$-n$	-14														
$3n$	$6n^6$	$-3n^2$	$-42n$														
2	$4n^5$	$-2n$	-28														
$6n^6 + 4n^5 - 3n^2 - 44n - 28$																	

- b. Compare your answers with your partner and resolve any differences.

Answers will vary.

- c. Discuss any connections you notice between the methods used to find the product.

Answers will vary. I noticed that they are doing the same multiplication but organized in a different way. I noticed that they result in the same answer but are structured differently.

- d. Complete the statement.

Answers will vary.

When multiplying polynomials, I prefer to use the

- ☐ **area model**
☐ **distributive property**

because ... *it is more*

organized, and I don't miss any calculations.



For question 2, students may ask for clarification on the directions. Each partner will use either the distributive property or the area model to complete the multiplication requested.



For physical movement, consider permitting students to find another student who used the same method and compare answers and steps prior to completing question 2a with their original partner.



This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

2.7.5 Your Turn!

Component Overview: Students will complete four independent practice questions involving multiplying polynomials using the method of their choice.



2.7.5 Your Turn!

- Find the product of each expression using your preferred method.

$\left(2x - \frac{1}{3}\right)\left(x + \frac{5}{4}\right)$ $2x^2 + \frac{13}{6}x - \frac{5}{12}$	$\left(\frac{2}{3} + a\right)(-5a + 2a^2 - 8)$ $2a^3 - \frac{11}{3}a^2 - \frac{34}{3}a - \frac{16}{3}$
$(7r - 0.18)(-6 + r^2 + 4r)$ $7r^3 + 27.82r^2 - 42.72r + 1.08$	$(2y + 1 + 30y^3)\left(-4 + y^3 + \frac{5}{4}y\right)$ $30y^6 + \frac{79}{2}y^4 - 119y^3 + \frac{5}{2}y^2 - \frac{27}{4}y - 4$



Consider allowing students the opportunity to use their calculators as part of this activity. Note the difference in students who are making procedural errors versus those who are simply making computational errors.



For students who appear to be struggling, consider writing an additional problem that uses whole numbers to see if they conceptually understand the steps to the multiplying process and need additional support with rational number operations.

2.7.6 Wrap-Up: Reflection

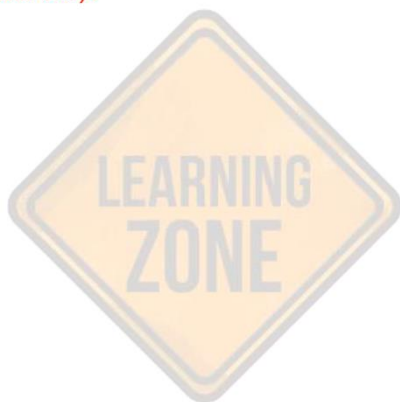
Component Overview: Students will complete an “in the zone” reflection where they describe how “in the zone” they feel with their understanding of multiplying polynomials. This activity should be done independently to determine student perception of the content from this lesson.



2.7.6 Wrap-Up: Reflection

1. How “in the zone” do you feel right now about multiplying polynomials?

Answers will vary.



Consider using student responses from prior self-reflections to consider a student’s learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.